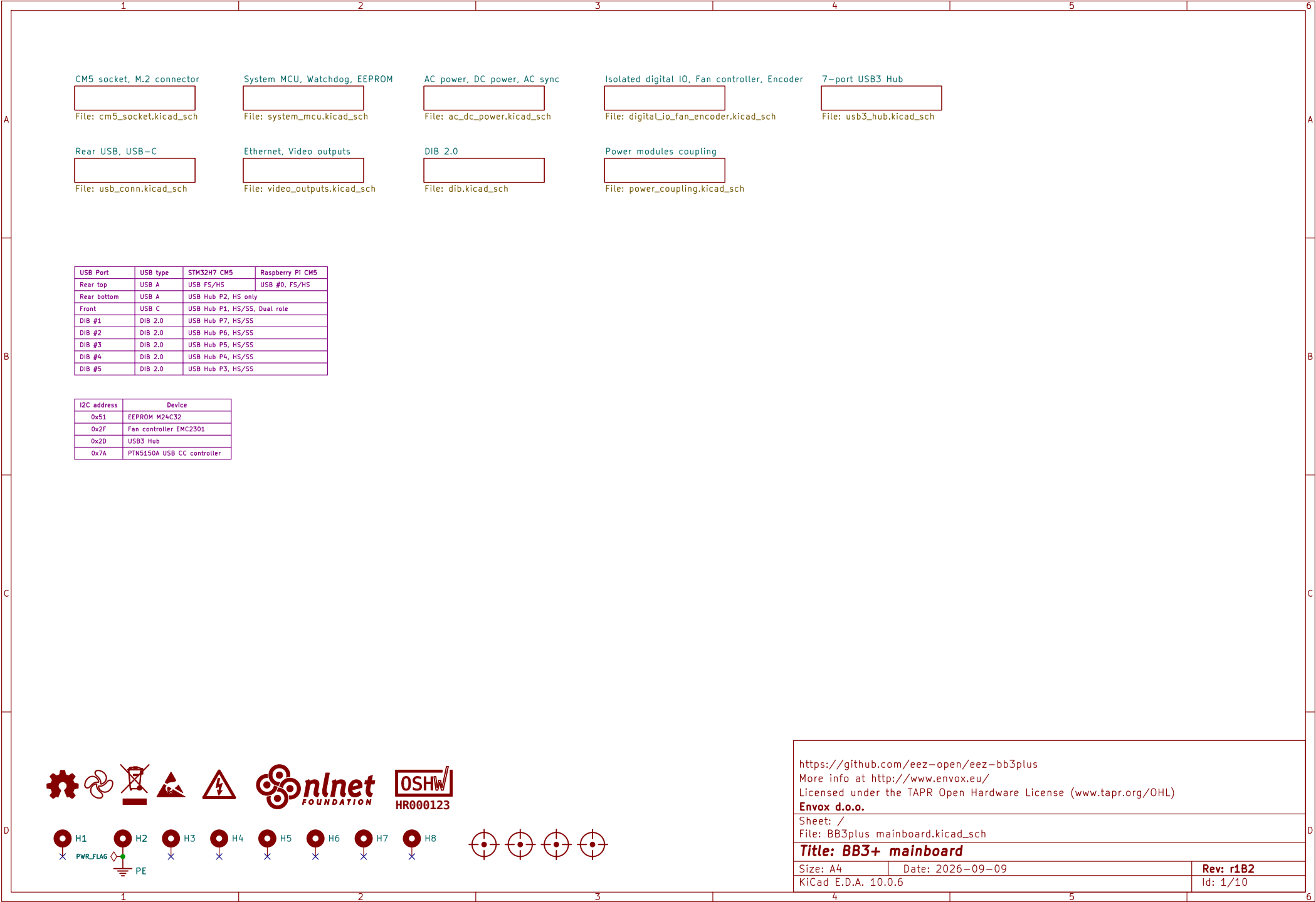
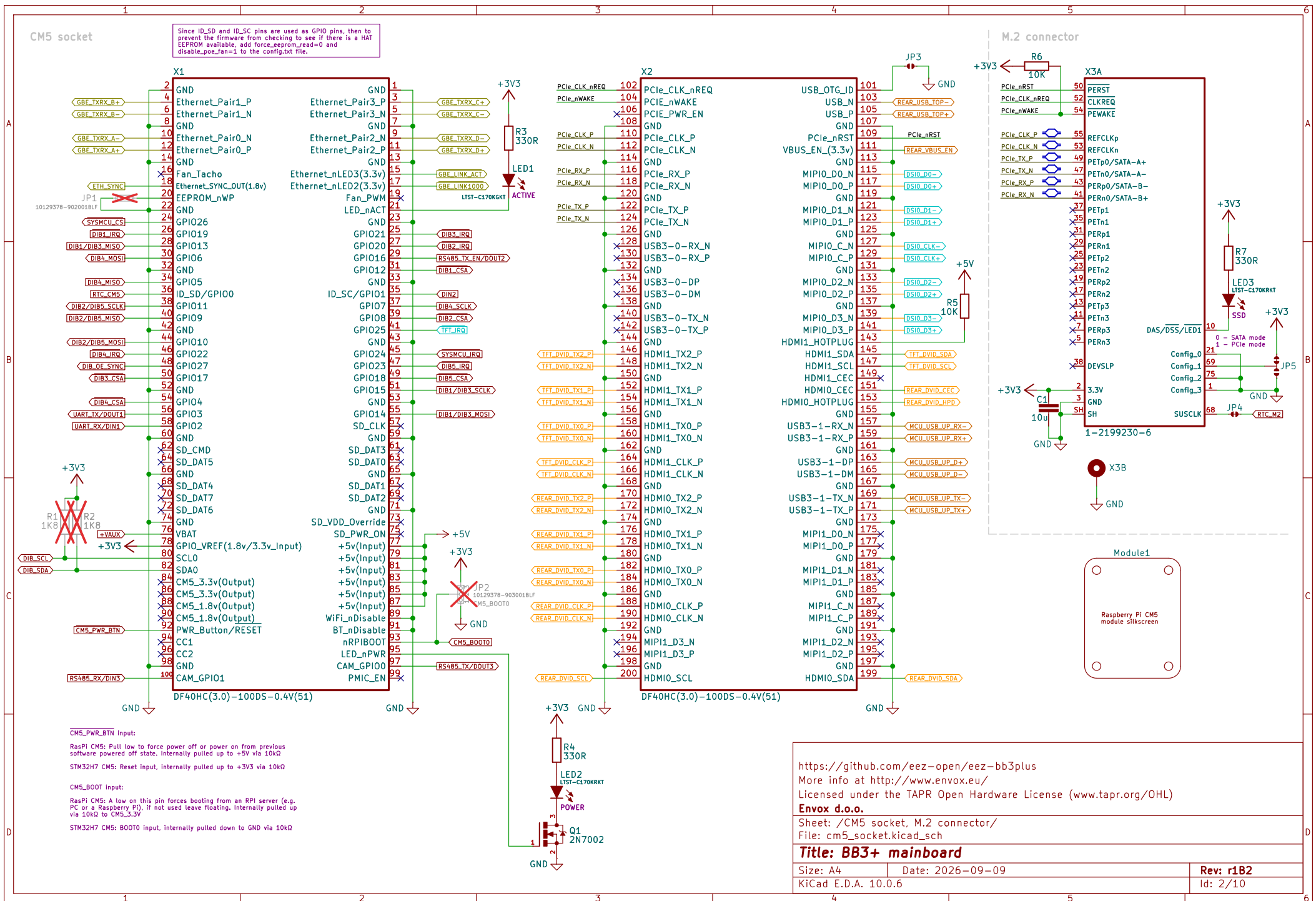
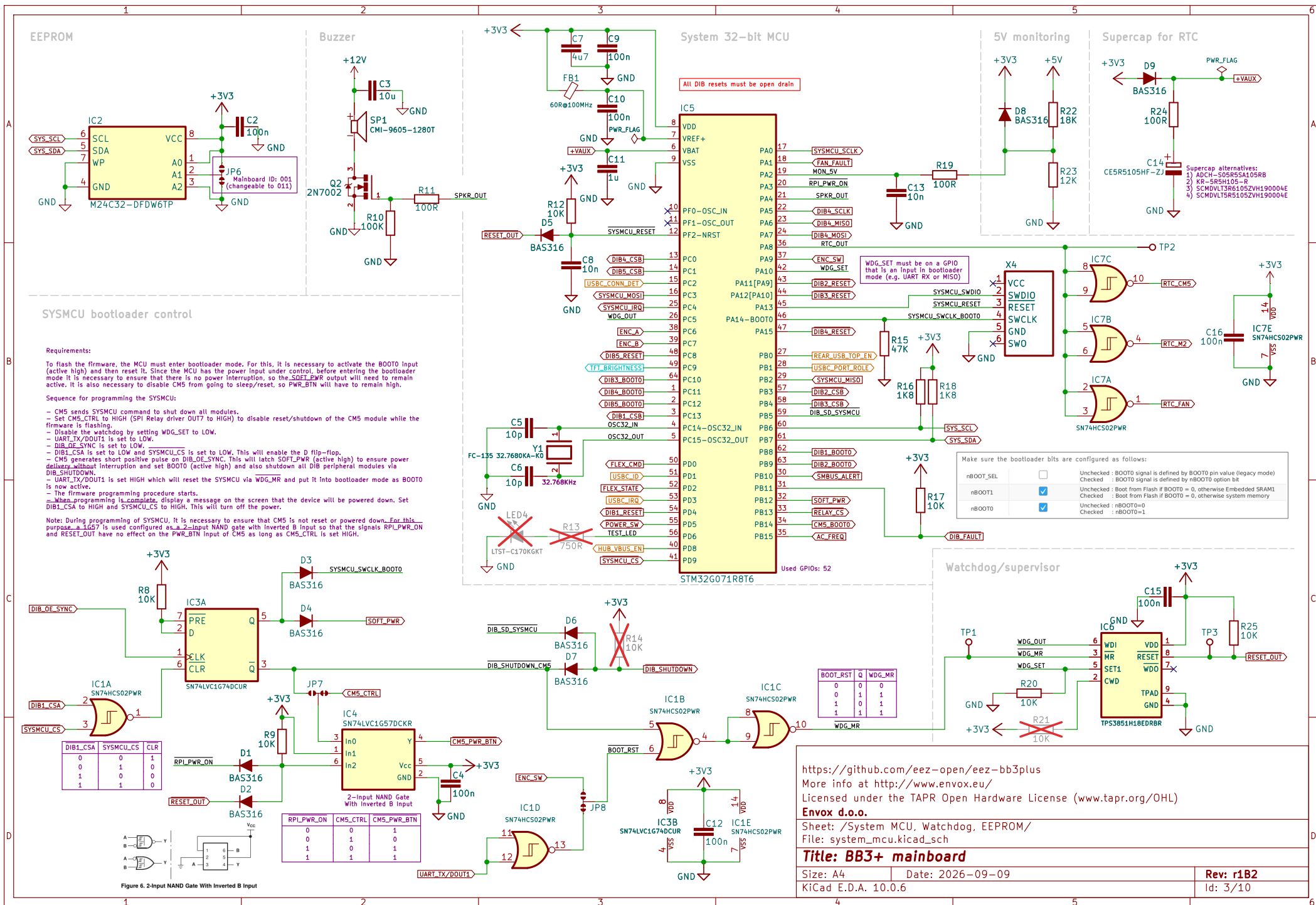
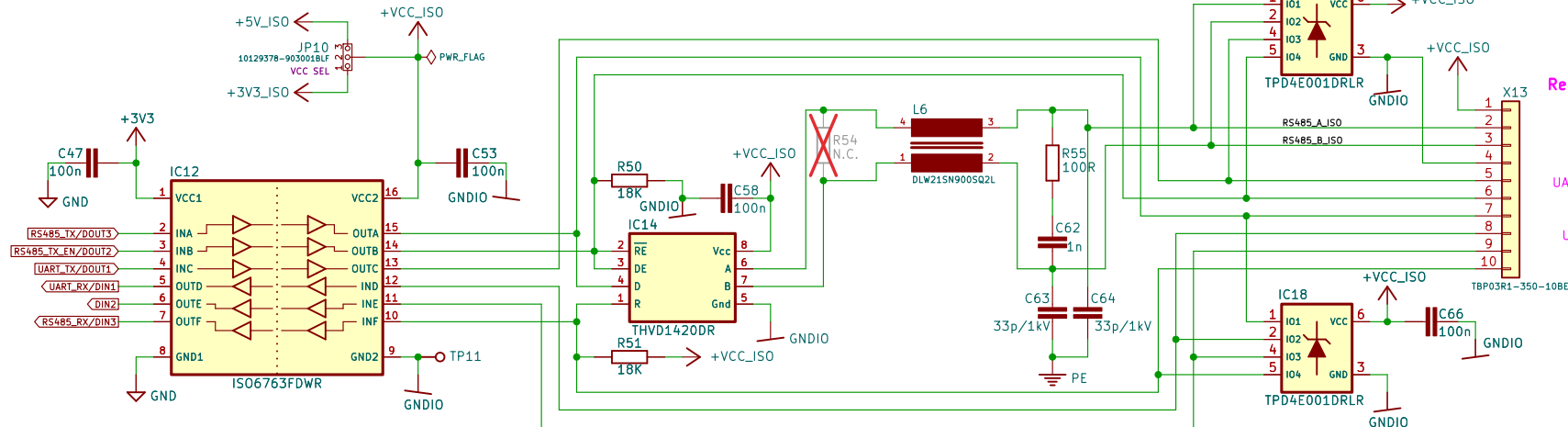




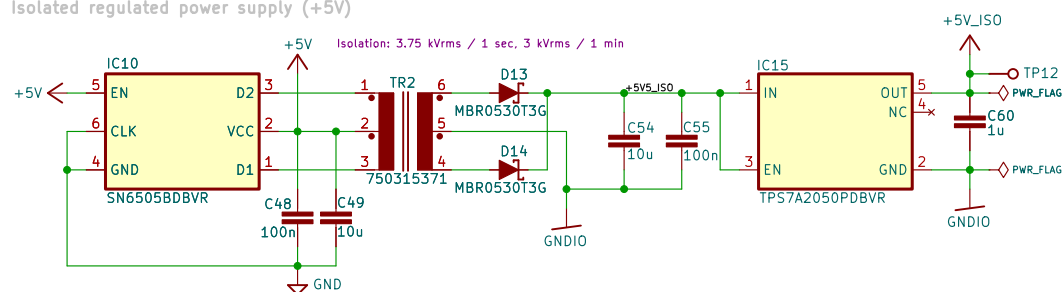




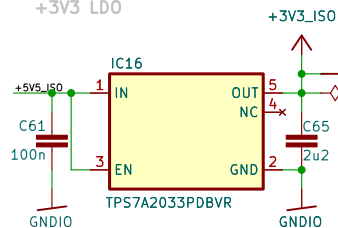
Isolated digital I/O & RS485 transceiver (half-duplex)



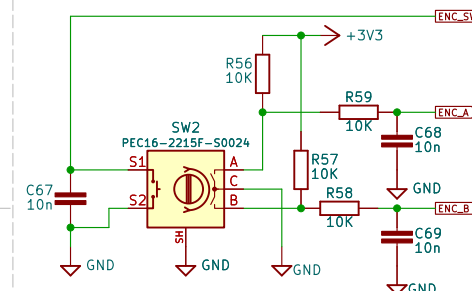
Isolated regulated power supply (+5V)



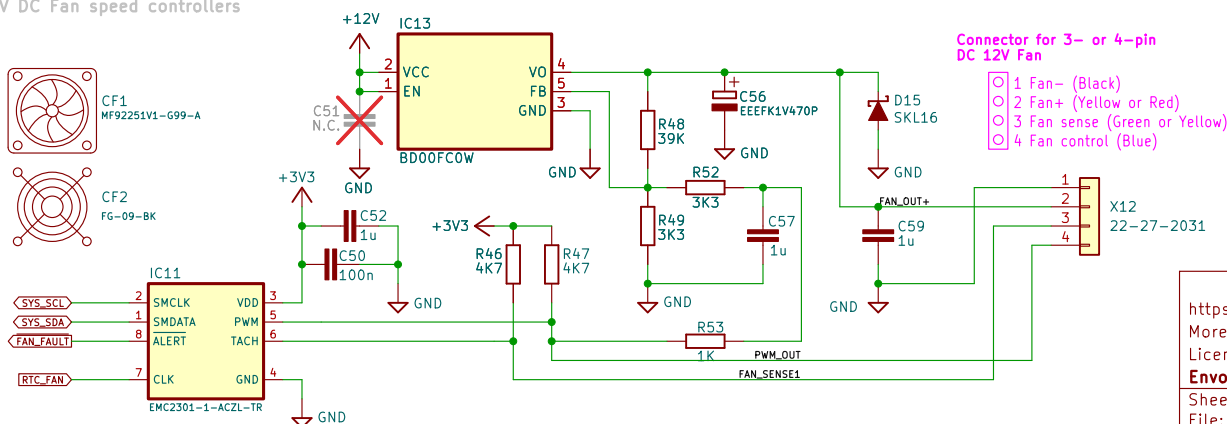
+3V3 LDO



Encoder with switch



12V DC Fan speed controllers



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

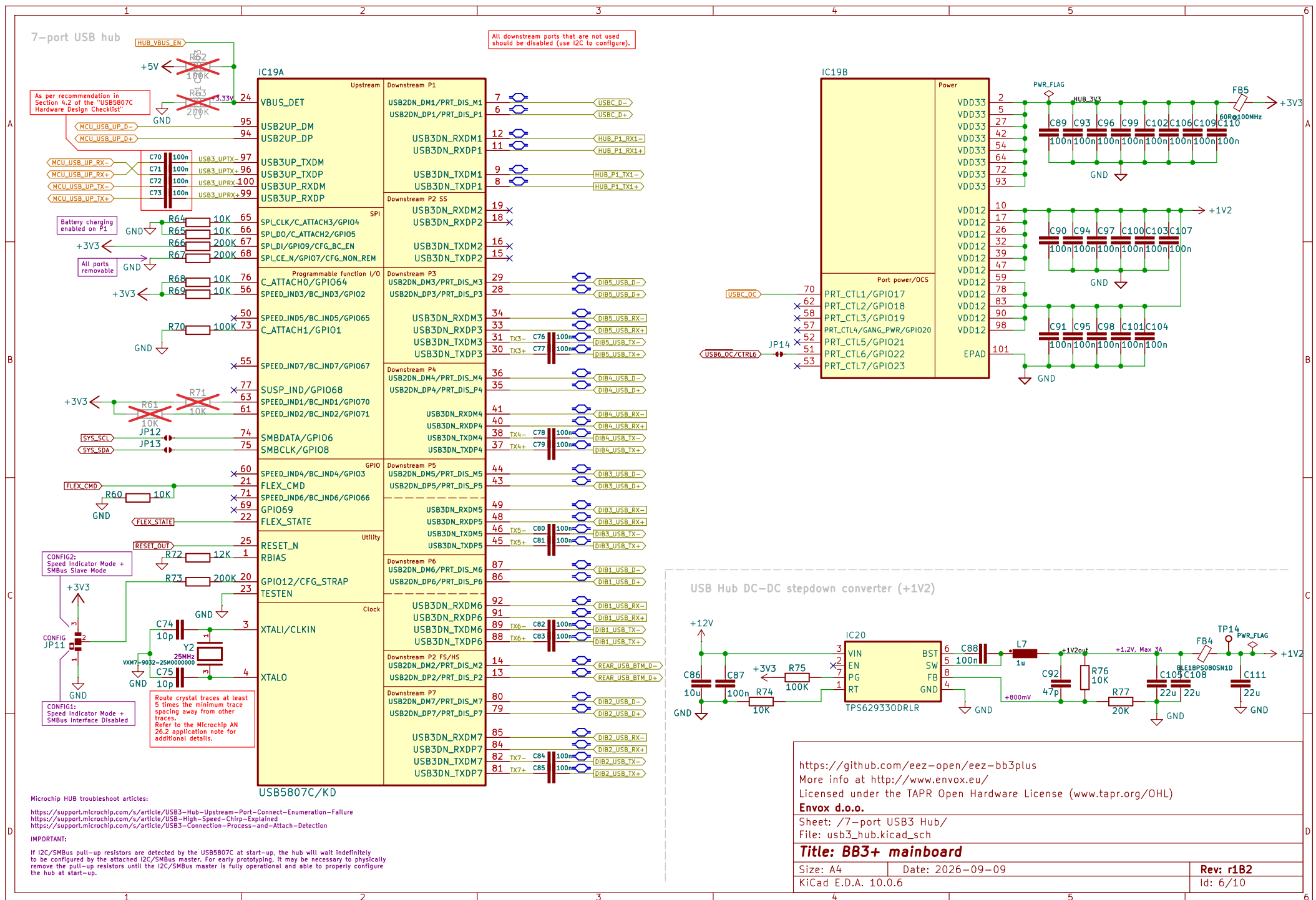
Title: BB3+ mainboard

Size: A4 Date: 2026-09-09

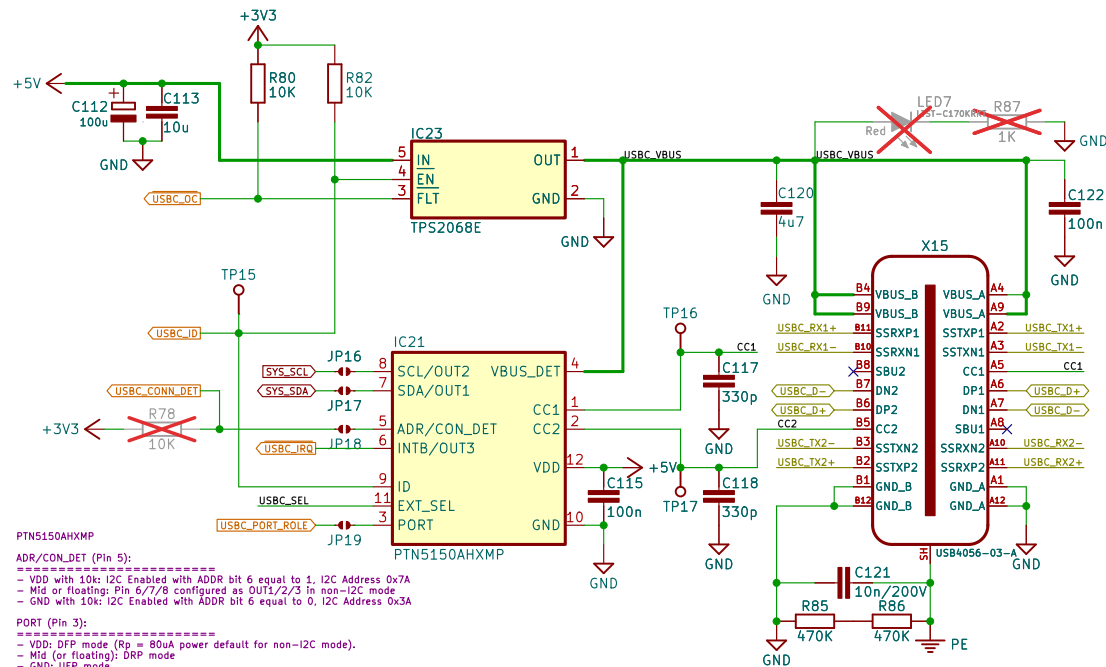
KiCad E.D.A. 10.0.6

Rev: r1B2

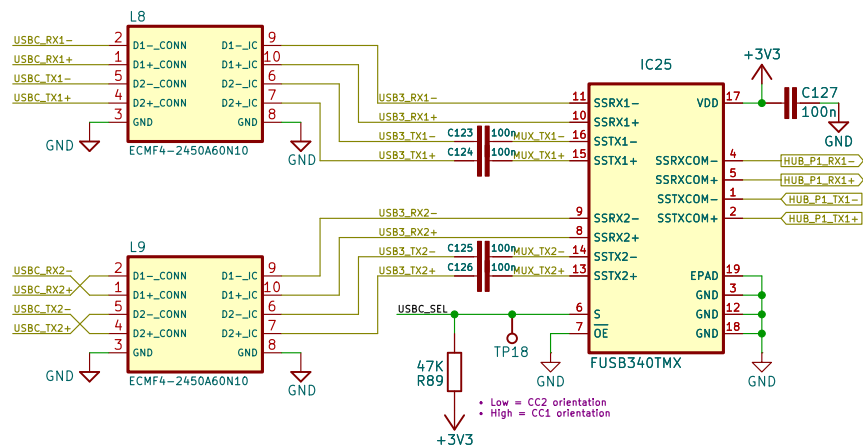
Id: 5/10



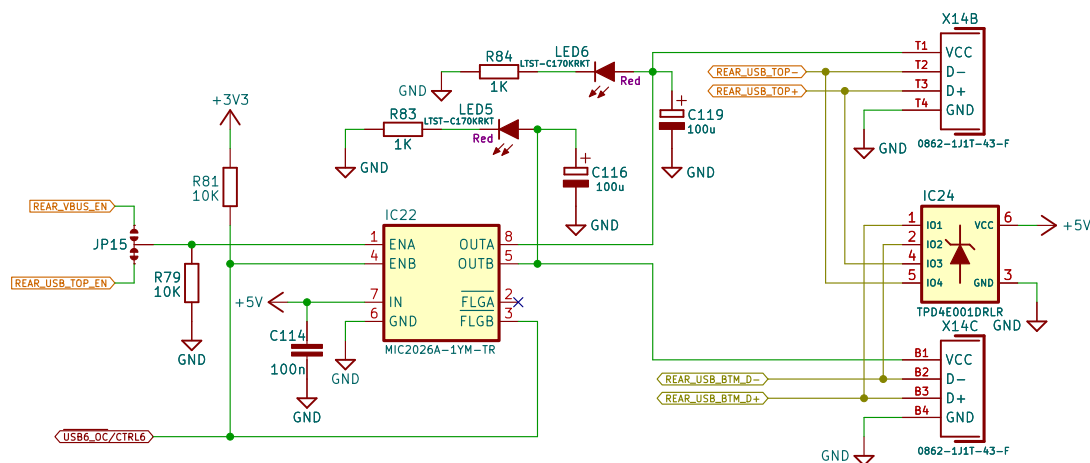
USB-C 3.0 front port (source max 1.5A)



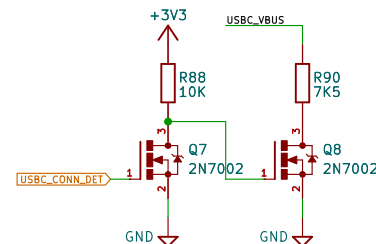
USB-C 3.0 Protection and Mux



USB2.0 Type-A (Rear panel)



VBus Discharge Circuit



<https://github.com/eez-open/eez-bb3plus>

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Sheet: /Rear USB, USB-C/

File: usb_conn.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-09-09

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Rev: r1B2

Id: 7/10

The schematic diagram illustrates the electrical connections for a rear display assembly. It features three main components: a display panel (X16, SS-53000-002), a TPS22948DCR IC (IC26), and two ECMA4-2450A60N10 connectors (L10 and L11).

Display Panel (X16) Connections:

- Power and Ground:** D2+ (1), D2S (2), D2- (3), D1+ (4), D1S (5), D1- (6), D0+ (7), D0S (8), D0- (9), CK+ (10), CKS (11), CK- (12), CEC (13), UTILITY/HEAC+ (14), SCL (15), SDA (16), GND (17), +5V (18), HPD/HEAC- (19).
- Control Lines:** REAR_DVID_CEC, REAR_DVID_SCL, REAR_DVID_SDA, REAR_DVID_HPD.

TPS22948DCR IC (IC26) Connections:

- Power and Ground:** IN (1), ON (2), FLT (3), OUT (4), GND (5).
- Control Lines:** REAR_DVID_HPD.

ECMA4-2450A60N10 Connectors (L10 and L11) Connections:

- Power and Ground:** D1-CONN (1), D1-JC (2), D1+CONN (3), D1-JC (4), D2-CONN (5), D2-JC (6), D2+CONN (7), D2-JC (8), GND (9).
- Control Lines:** REAR_DVID_CEC, REAR_DVID_SCL, REAR_DVID_SDA, REAR_DVID_HPD.

Other Components and Connections:

- Resistors:** R93 (4K7), R94 (470K), R95 (470K).
- Capacitors:** C128 (100n), C130 (10u), C131 (100n), C132 (10n/200V).
- Power Sources:** +5V, +3V3, PE.

The diagram illustrates the electrical connections for the X14A module. It features three main connector blocks: L12, L13, and L14. L12 and L13 are populated with ECMF4-2450A60N10 components, while L14 is populated with an 0862-1J1T-43-F component. The connections include:

- GBE_TXRX_C+/-** and **GBE_TXRX_A+/-** signals connected to L12 pins 1, 2, 3, 4, 5, and 6.
- GBE_TXRX_D+/-** and **GBE_TXRX_B+/-** signals connected to L13 pins 1, 2, 3, 4, 5, and 6.
- GBE_LINK_ACT** and **GBE_LINK1000** signals connected to L14 pins 13 and 14.
- Power and Ground** connections: +3V3, GND, and a 60R@100MHz component are connected to various pins on L12, L13, and L14.
- Other Connections**: TRD3-, TRD3+, TRD1-, TRD1+, TRD2-, TRD2+, and COMMON_CT are connected to L12 and L13 pins.

The diagram also shows a 3V3 power supply connected to a 60R@100MHz component, which is connected to the SHIELD pin of L14. A 60R@100MHz component is also connected to the SHIELD pin of L14. The diagram includes a 60R@100MHz component and a 60R@100MHz component.

The diagram shows the pin connections for the X17 505110-2291 module. The module is represented by a red box with pins numbered 1 to 22. The connections are as follows:

- Pin 1:** GND
- Pin 2:** DPHY_CLK_N
- Pin 3:** DPHY_CLK_P
- Pin 4:** GND
- Pin 5:** DPHY_D0_N
- Pin 6:** DPHY_D0_P
- Pin 7:** GND
- Pin 8:** DPHY_D1_N
- Pin 9:** DPHY_D1_P
- Pin 10:** GND
- Pin 11:** DPHY_D2_N
- Pin 12:** DPHY_D2_P
- Pin 13:** GND
- Pin 14:** EDID_SCL
- Pin 15:** EDID_SDA
- Pin 16:** GND
- Pin 17:** GPIO0/BL_PWM
- Pin 18:** GPIO1/TOUCH_IRQ
- Pin 19:** GND
- Pin 20:** SCL
- Pin 21:** SDA
- Pin 22:** +VCC

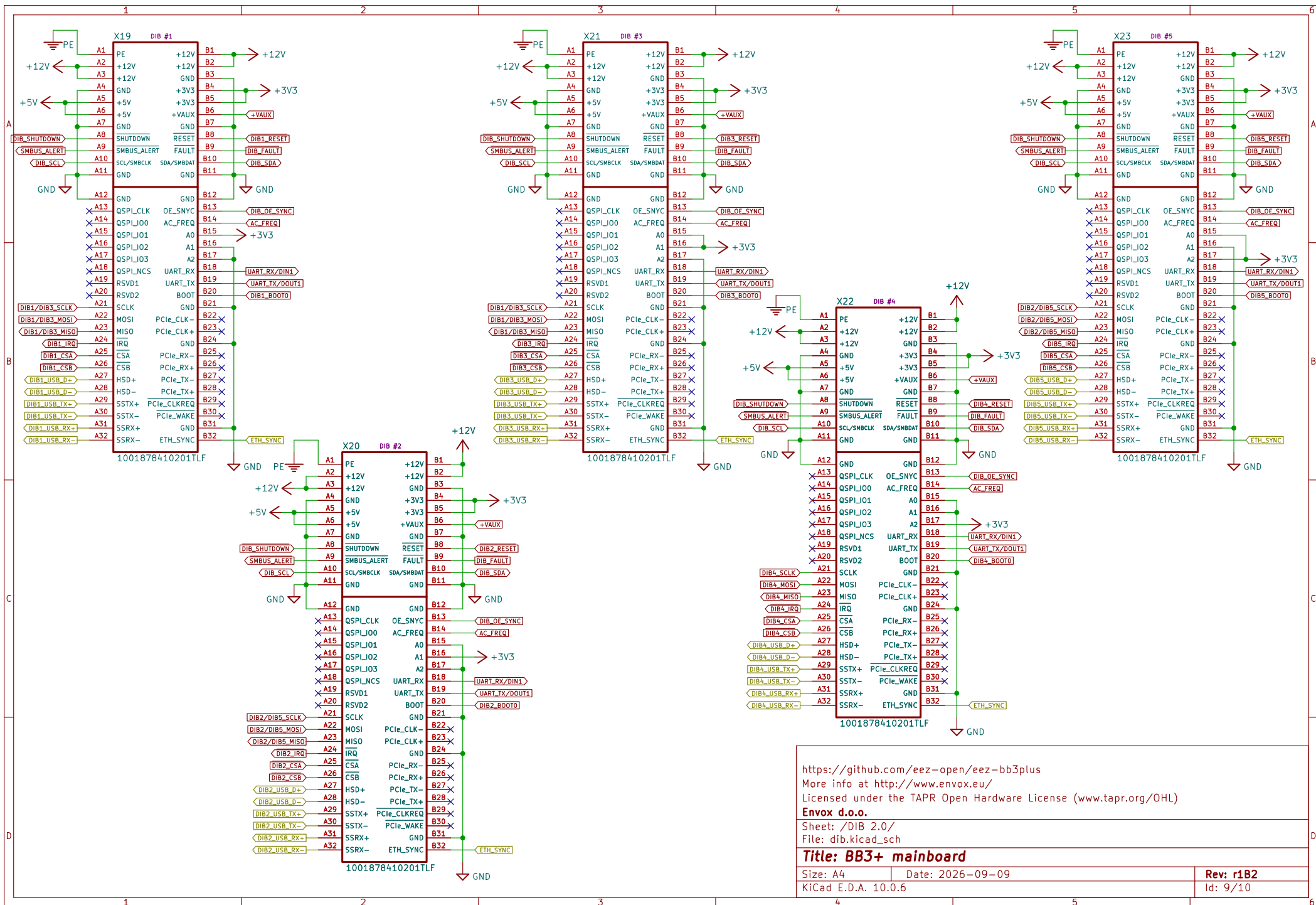
External components and connections include:

- JP20:** A 2-pin header connected to pins 17 and 18.
- PWR_FLAG:** A push-button switch connected to pin 22 and GND.
- Power:** +5V and +3V3 power sources connected to the module's power pins.
- Grounding:** Multiple GND connections are shown, including a common GND at the bottom.

Title: BB3+ mainboard

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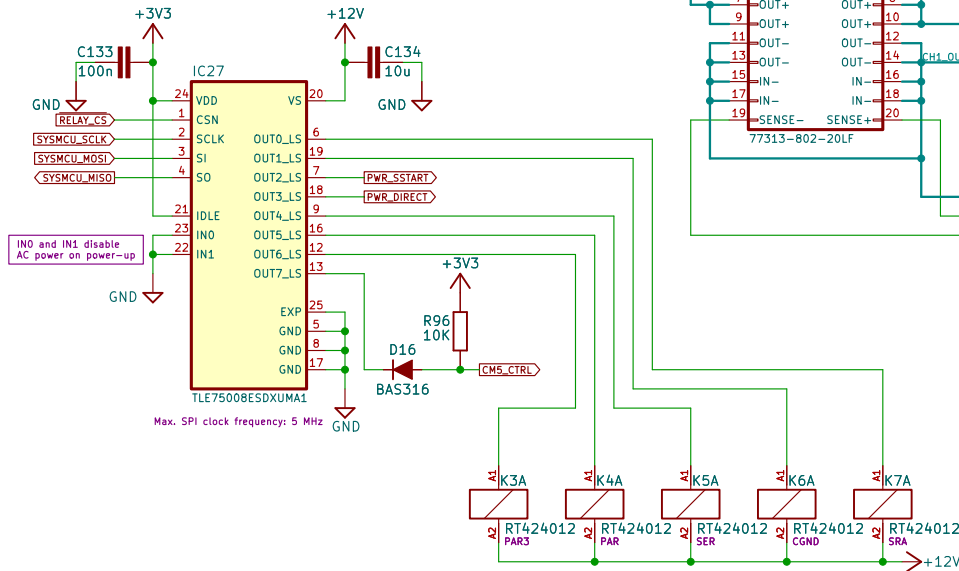
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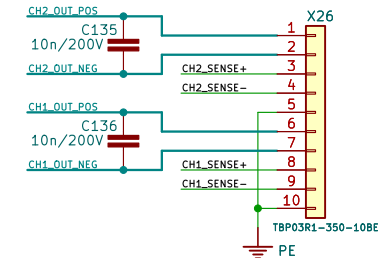
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	2 PWR+ IN
PWR+ IN 3	4 PWR+ IN
PWR+ IN 5	6 PWR+ OUT
PWR+ OUT 7	8 PWR+ OUT
PWR+ OUT 9	10 PWR+ OUT
PWR- OUT 11	12 PWR- OUT
PWR- OUT 13	14 PWR- OUT
PWR- IN 15	16 PWR- IN
PWR- IN 17	18 PWR- IN
SENSE- 19	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+ 1	2
CH2_OUT- 3	4
CH2_SENSE+ 5	6
CH2_SENSE- 7	8
PE 9	10
CH1_OUT+ 11	12
CH1_OUT- 13	14
CH1_SENSE+ 15	16
CH1_SENSE- 17	18
PE 19	20

<https://github.com/eez-open/eez-bb3plus>

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Sheet: //Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-09-09

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Rev: r1B2

Id: 10/10